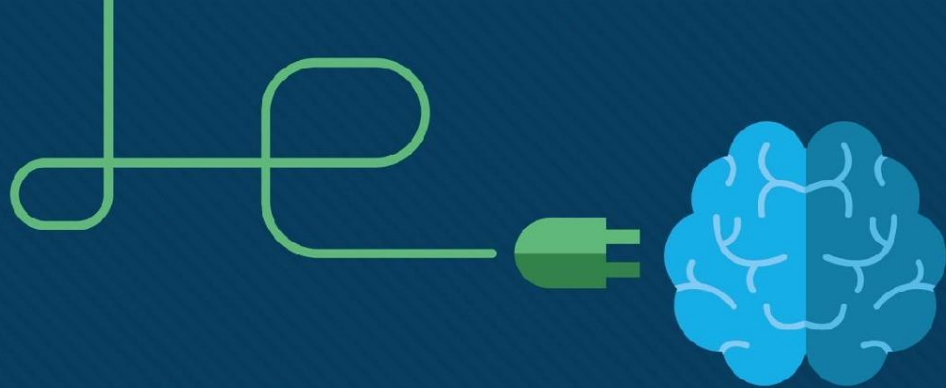




Chapter 1 Part 1

Module 15: IP Static Routing

Switching, Routing, and Wireless Essentials v7.0
(SRWE)



Module Objectives

Module Title: IP Static Routing

Module Objective: Configure IPv4 and IPv6 static routes.

Topic Title	Topic Objective
Static Routes	Describe the command syntax for static routes.
Configure IP Static Routes	Configure IPv4 and IPv6 static routes.
Configure IP Default Static Routes	Configure IPv4 and IPv6 default static routes.
Configure Floating Static Routes	Configure a floating static route to provide a backup connection.
Configure Static Host Routes	Configure IPv4 and IPv6 static host routes that direct traffic to a specific host.

15.1 Static Routes

Types of Static Routes

Static routes are commonly implemented on a network. This is true even when there is a dynamic routing protocol configured.

Static routes can be configured for IPv4 and IPv6. Both protocols support the following types of static routes:

- Standard static route
- Default static route
- Floating static route
- Summary static route

Static routes are configured using the **ip route** and **ipv6 route** global configuration commands.

Static Routes

Next-Hop Options

When configuring a static route, the next hop can be identified by an **IP address**, **exit interface**, or **both**. How the destination is specified creates one of the three following types of static route:

- **Next-hop static route** - Only the **next-hop IP address** is specified
- **Directly connected static route** - Only the router **exit interface** is specified
- **Fully specified static route** - The **next-hop IP address** and **exit interface** are specified

IPv4 Static Route Command

IPv4 static routes are configured using the following global configuration command:

```
Router(config)# ip route network-address subnet-mask { ip-address  
| exit-intf [ip-address] } [distance]
```

Note: Either the *ip-address*, *exit-intf*, or the *ip-address* and *exit-intf* parameters must be configured.

IPv6 Static Route Command

IPv6 static routes are configured using the following global configuration command:

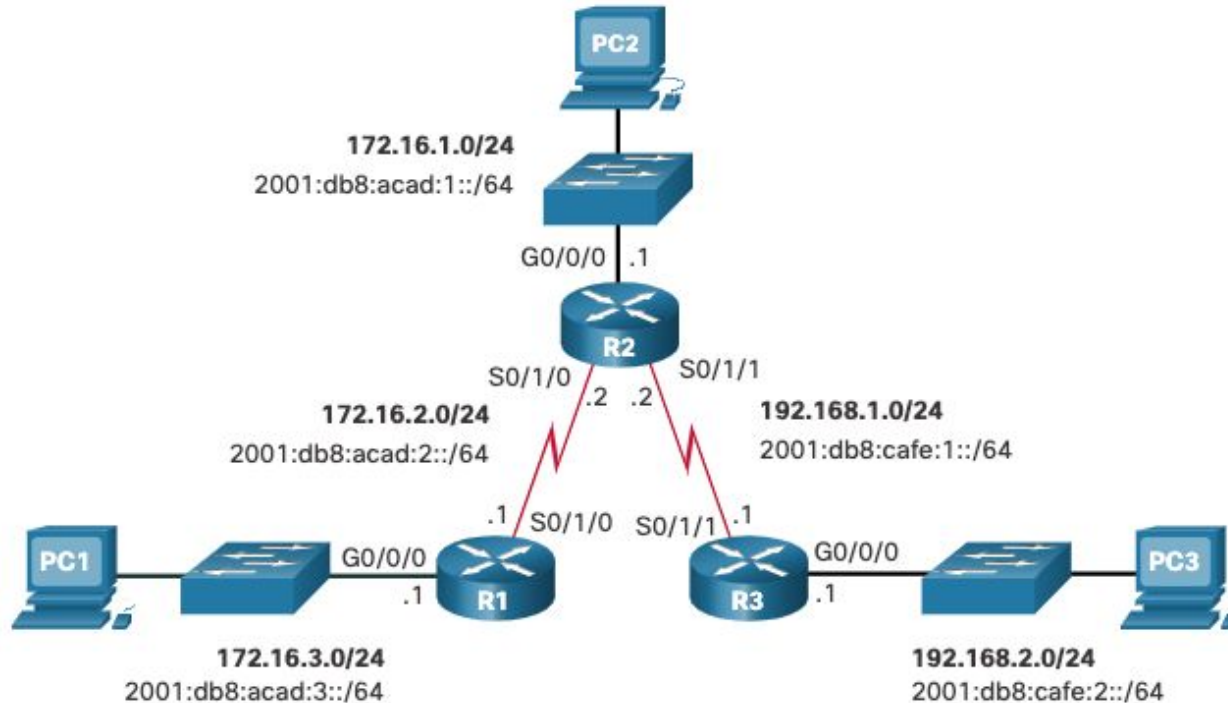
```
Router(config)# ipv6 route ipv6-prefix/prefix-length {ipv6-address  
| exit-intf [ipv6-address]} [distance]
```

Most of parameters are identical to the IPv4 version of the command.

Static Routes

Dual-Stack Topology

The figure shows a dual-stack network topology. Currently, no static routes are configured for either IPv4 or IPv6.



Static Routes

IPv4 Starting Routing Tables

Each router has entries only for directly connected networks and associated local addresses.

R1 can ping R2, but cannot ping the R3 LAN

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.1/32 is directly connected, Serial0/1/0
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0/0
R1#
R1# ping 172.16.2.2
Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 172.16.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5)
R1# ping 192.168.2.1
Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Static Routes

IPv6 Starting Routing Tables

Each router has entries only for directly connected networks and associated local addresses.
R1 can ping R2, but cannot ping the R3 LAN.

```
R1# show ipv6 route | begin C
C 2001:DB8:ACAD:2::/64 [0/0]
    via Serial0/1/0, directly connected
L 2001:DB8:ACAD:2::1/128 [0/0]
    via Serial0/1/0, receive
C 2001:DB8:ACAD:3::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
L FF00::/8 [0/0]
    via Null0, receive

R1#
R1# ping 2001:db8:acad:2::2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:2::2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
R1# ping 2001:DB8:cafe:2::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:CAFE:2::1, timeout is 2 seconds:
% No valid route for destination
Success rate is 0 percent (0/1)
```

15.2 Configure IP Static Routes

Configure IP Static Routes

IPv4 Next-Hop Static Route

In a next-hop static route, only the next-hop IP address is specified. The exit interface is derived from the next hop. For example, three next-hop IPv4 static routes are configured on R1 using the IP address of the next hop, R2.

```
R1(config)# ip route 172.16.1.0 255.255.255.0 172.16.2.2
```

```
R1(config)# ip route 192.168.1.0 255.255.255.0 172.16.2.2
```

```
R1(config)# ip route 192.168.2.0 255.255.255.0 172.16.2.2
```

The resulting routing table entries on R1:

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.16.1.0/24 [1/0] via 172.16.2.2
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.1/32 is directly connected, Serial0/1/0
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0/0
S       192.168.1.0/24 [1/0] via 172.16.2.2
S       192.168.2.0/24 [1/0] via 172.16.2.2
```

Configure IP Static Routes

IPv6 Next-Hop Static Route

The commands to configure R1 with the IPv6 static routes to the three remote networks are as follows:

```
R1(config)# ipv6 unicast-routing
```

```
R1(config)# ipv6 route 2001:db8:acad:1::/64  
2001:db8:acad:2::2
```

```
R1(config)# ipv6 route 2001:db8:cafe:1::/64  
2001:db8:acad:2::2
```

```
R1(config)# ipv6 route 2001:db8:cafe:2::/64  
2001:db8:acad:2::2
```

The routing table for R1 now has routes to the three remote IPv6 networks.

```
R1# show ipv6 route
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       a - Application
S 2001:DB8:ACAD:1::/64 [1/0]
  via 2001:DB8:ACAD:2::2
C 2001:DB8:ACAD:2::/64 [0/0]
  via Serial0/1/0, directly connected
L 2001:DB8:ACAD:2::1/128 [0/0]
  via Serial0/1/0, receive
C 2001:DB8:ACAD:3::/64 [0/0]
  via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
  via GigabitEthernet0/0/0, receive
S 2001:DB8:CAFE:1::/64 [1/0]
  via 2001:DB8:ACAD:2::2
S 2001:DB8:CAFE:2::/64 [1/0]
  via 2001:DB8:ACAD:2::2
L FF00::/8 [0/0]
  via Null0, receive
```

IPv4 Directly Connected Static Route

When configuring a static route, another option is to use the exit interface to specify the next-hop address. Three directly connected IPv4 static routes are configured on R1 using the exit interface.

Note: Using a next-hop address is generally recommended. Directly connected static routes should only be used with point-to-point serial interfaces.

```
R1(config)# ip route 172.16.1.0 255.255.255.0 s0/1/0
```

```
R1(config)# ip route 192.168.1.0 255.255.255.0 s0/1/0
```

```
R1(config)# ip route 192.168.2.0 255.255.255.0 s0/1/0
```

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.16.1.0/24 is directly connected, Serial0/1/0
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.1/32 is directly connected, Serial0/1/0
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0/0
S       192.168.1.0/24 is directly connected, Serial0/1/0
S       192.168.2.0/24 is directly connected, Serial0/1/0
```

Configure IP Static Routes

IPv6 Directly Connected Static Route

In the example, three directly connected IPv6 static routes are configured on R1 using the exit interface.

Note: Using a next-hop address is generally recommended. Directly connected static routes should only be used with point-to-point serial interfaces.

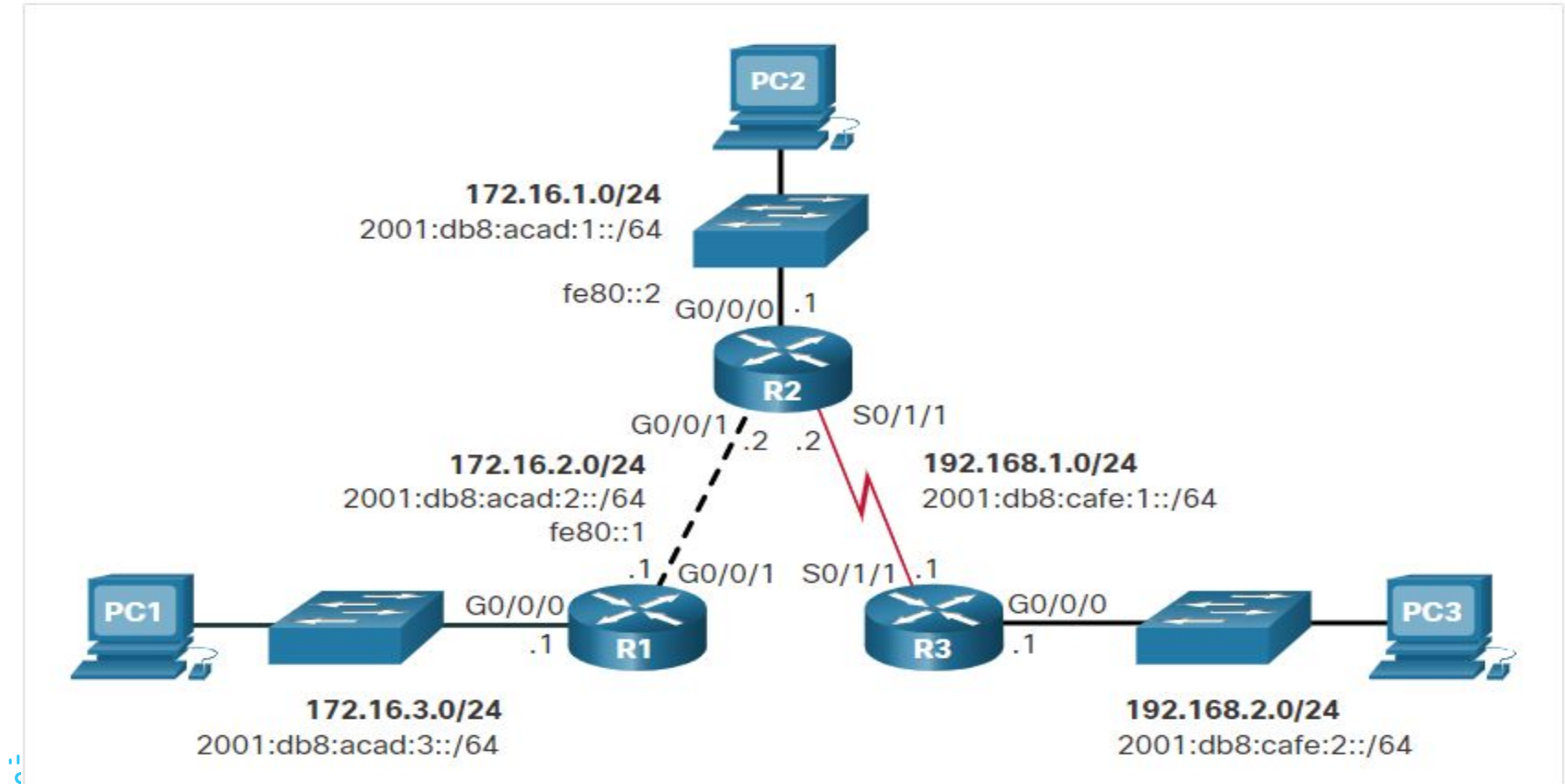
```
R1(config)# ipv6 route 2001:db8:acad:1::/64  
s0/1/0
```

```
R1(config)# ipv6 route 2001:db8:cafe:1::/64  
s0/1/0
```

```
R1(config)# ipv6 route 2001:db8:cafe:2::/64  
s0/1/0
```

```
R1# show ipv6 route
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       a - Application
S 2001:DB8:ACAD:1::/64 [1/0]
  via Serial0/1/0, directly connected
C 2001:DB8:ACAD:2::/64 [0/0]
  via Serial0/1/0, directly connected
L 2001:DB8:ACAD:2::1/128 [0/0]
  via Serial0/1/0, receive
C 2001:DB8:ACAD:3::/64 [0/0]
  via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
  via GigabitEthernet0/0/0, receive
S 2001:DB8:CAFE:1::/64 [1/0]
  via Serial0/1/0, directly connected
S 2001:DB8:CAFE:2::/64 [1/0]
  via Serial0/1/0, directly connected
L FF00::/8 [0/0]
  via Null0, receive
IPv6 Routing Table - default - 8 entries
R1#
```

IPv4 Fully Specified Static Route



Configure IP Static Routes

IPv4 Fully Specified Static Route

In a fully specified static route, **both the exit interface and the next-hop IP address are specified**. This form of static route is used when the **exit interface is a multi-access interface** and it is necessary to explicitly identify the next hop. The next hop must be directly connected to the specified exit interface. Using an exit interface is optional, however it is necessary to use a next-hop address.

It is recommended that when the exit interface is an **Ethernet network**, that the static route includes a next-hop address. You can also use a fully specified static route that includes both the exit interface and the next-hop address.

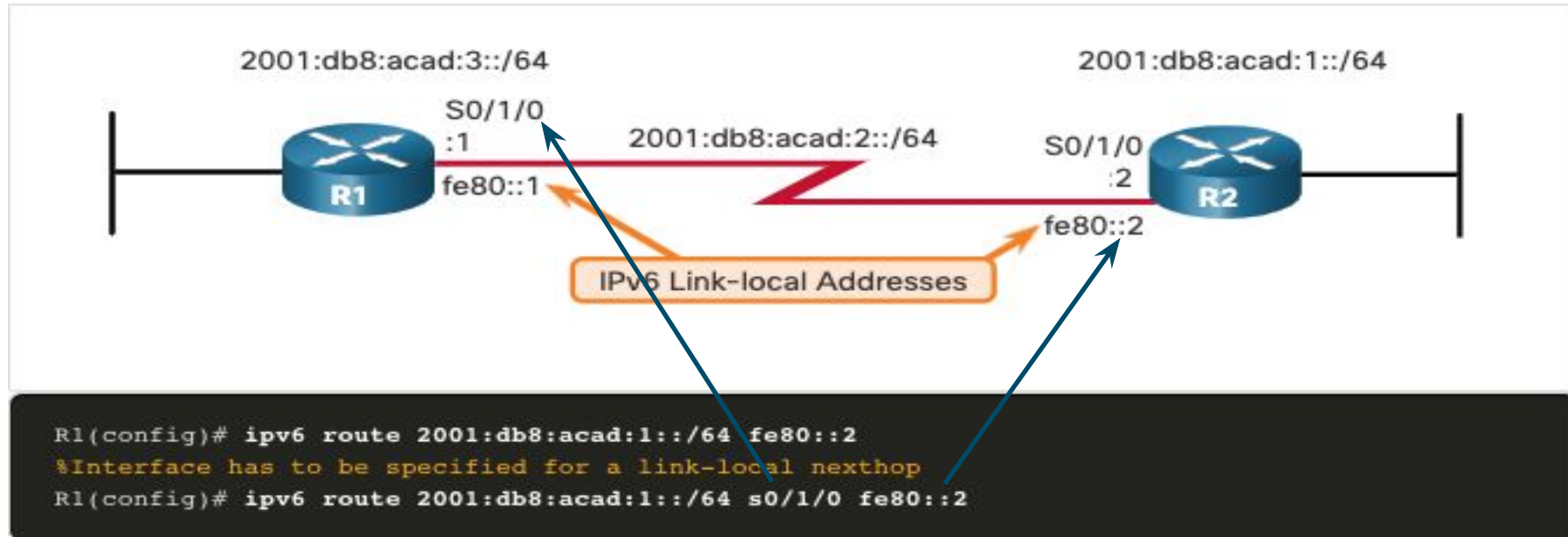
```
R1(config)# ip route 172.16.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
R1(config)# ip route 192.168.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
R1(config)# ip route 192.168.2.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
```

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.16.1.0/24 [1/0] via 172.16.2.2, GigabitEthernet0/0/1
C       172.16.2.0/24 is directly connected, GigabitEthernet0/0/1
L       172.16.2.1/32 is directly connected, GigabitEthernet0/0/1
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0/0
S      192.168.1.0/24 [1/0] via 172.16.2.2, GigabitEthernet0/0/1
S      192.168.2.0/24 [1/0] via 172.16.2.2, GigabitEthernet0/0/1
```

Configure IP Static Routes

IPv6 Fully Specified Static Route

In a fully specified static route, **both the exit interface and the next-hop IPv6 address** are specified. There is a situation in IPv6 when a fully specified static route **must** be used. If the IPv6 static route uses an **IPv6 link-local address** as the **next-hop address**, use **a fully specified static route**. The figure shows an example of a fully specified IPv6 static route using an IPv6 link-local address as the next-hop address.



IPv6 Fully Specified Static Route (Cont.)

The reason a fully specified static route must be used is because **IPv6 link-local addresses** are **not contained in the IPv6 routing table**. Link-local addresses are only unique on a given link or network. The next-hop link-local address may be a valid address on multiple networks connected to the router. Therefore, it is necessary that the exit interface be included.

The following example shows the IPv6 routing table entry for this route. Notice that both the **next-hop link-local address** and **the exit interface** are included.

```
R1# show ipv6 route static | begin 2001:db8:acad:1::/64
S    2001:DB8:ACAD:1::/64 [1/0]
    via FE80::2, Serial0/1/0
```

Configure IP Static Routes

Verify a Static Route

Along with **show ip route**, **show ipv6 route**, **ping** and **tracert**, other useful commands to verify static routes include the following:

show ip route static

show ip route *network*

show running-config | section ip route

Replace **ip** with **ipv6** for the IPv6 versions of the command.

15.3 Configure IP Default Static Routes

Configure IP Default Static Routes

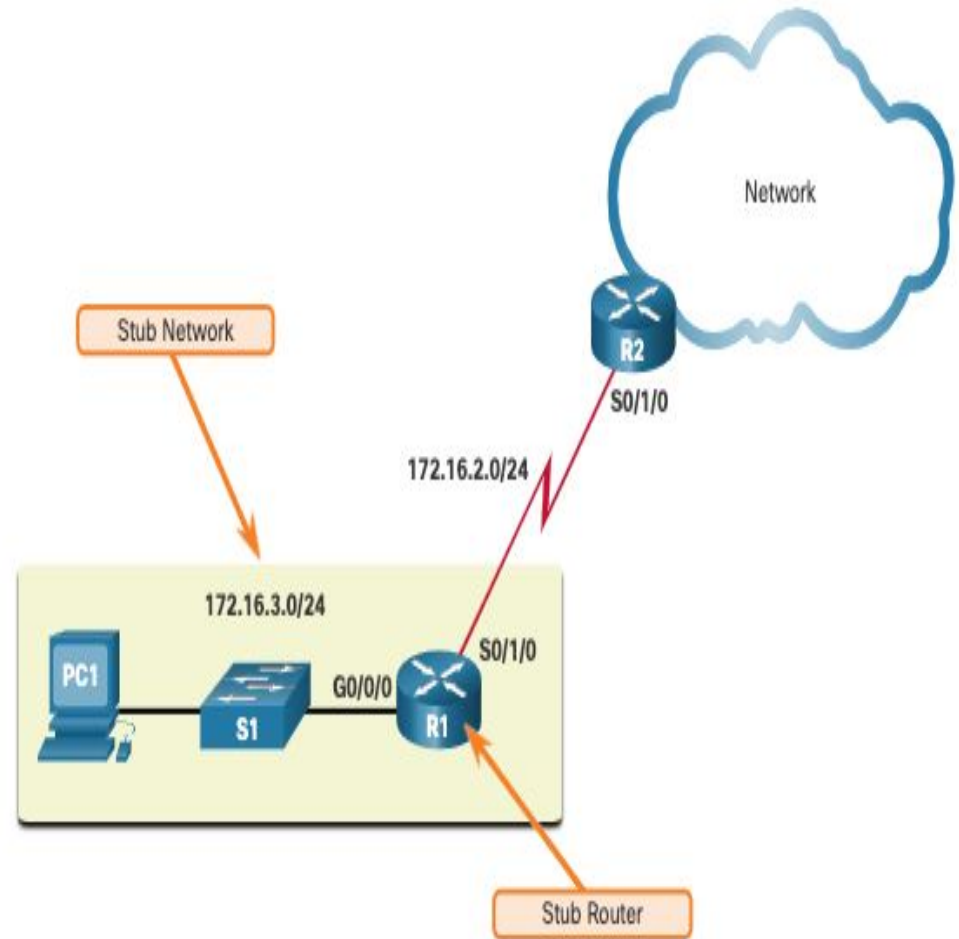
Default Static Route

A **default route** is a static route that **matches all** packets. A single default route represents **any network** that is not in the routing table.

Routers commonly use default routes that are either configured locally or learned from another router. The default route is used as the **Gateway of Last Resort**.

Default static routes are commonly used when connecting an **edge router** to a service provider network, or a stub router (a router with only one upstream neighbor router).

The figure shows a typical default static route scenario.



Configure IP Default Static Routes

Default Static Route (Cont.)

IPv4 Default Static Route: The command syntax for an IPv4 default static route is similar to any other IPv4 static route, except that the **network address is 0.0.0.0** and the **subnet mask is 0.0.0.0**. The 0.0.0.0 0.0.0.0 in the route will match any network address.

Note: An IPv4 default static route is commonly referred to as a quad-zero route.

The basic command syntax for an IPv4 default static route is as follows:

```
Router(config)# ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf}
```

IPv6 Default Static Route: The command syntax for an IPv6 default static route is similar to any other IPv6 static route, except that the ipv6-prefix/prefix-length is **::/0**, which matches all routes.

The basic command syntax for an IPv6 default static route is as follows:

```
Router(config)# ipv6 route ::/0 {ipv6-address | exit-intf}
```

Configure a Default Static Route

The example shows an IPv4 default static route configured on R1. With the configuration shown in the example, any packets not matching more specific route entries are forwarded to R2 at 172.16.2.2.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.2.2
```

An IPv6 default static route is configured in similar fashion. With this configuration any packets not matching more specific IPv6 route entries are forwarded to R2 at 2001:db8:acad:2::2

```
R1(config)# ipv6 route ::/0 2001:db8:acad:2::2
```

Configure IP Default Static Routes

Verify a Default Static Route

The **show ip route static** command output from R1 displays the contents of the static routes in the routing table. Note the asterisk (*) next to the route with code 'S'. The asterisk indicates that this static route is a **candidate default route**, which is why it is selected as the **Gateway of Last Resort**.

Notice that the static default route configuration uses the **/0 mask** for IPv4 default routes. Remember that the IPv4 subnet mask in a routing table determines how many bits must match between the destination IP address of the packet and the route in the routing table. **A /0 mask indicates that none of the bits are required to match.** As long as a more specific match does not exist, the default static route matches all packets.

```
R1# show ip route static
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is 172.16.2.2 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 172.16.2.2
```

Configure IP Default Static Routes

Verify a Default Static Route (Cont.)

This example shows the **show ipv6 route static** command output to display the contents of the routing table.

Notice that the static default route configuration uses the **::/0** prefix for IPv6 default routes. Remember that the IPv6 prefix-length in a routing table determines how many bits must match between the destination IP address of the packet and the route in the routing table. A **::/0** prefix indicates that none of the bits are required to match. As long as a more specific match does not exist, the default static route matches all packets.

```
R1# show ipv6 route static
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, la - LISP away, le - LISP extranet-policy
       a - Application
S      ::/0 [1/0]
      via 2001:DB8:ACAD:2::2
```

15.4 Configure Floating Static Routes

Configure Floating Static Routes

Floating Static Routes

Another type of static route is a **floating static route**. Floating static routes are static routes that are used to provide **a backup path** to a **primary static or dynamic route**. The floating static route is only used when the primary route is not available.

To accomplish this, the floating static route is configured with a **higher administrative distance** than the **primary route**. The administrative distance represents the trustworthiness of a route. If **multiple paths** to the destination exist, the router will choose the path with the **lowest administrative distance**.

By default, static routes have an administrative distance of **1**, making them preferable to routes learned from dynamic routing protocols.

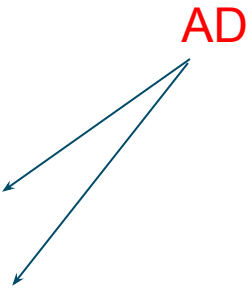
The administrative distance of a static route can be increased to make the route less desirable than that of another static route or a route learned through a dynamic routing protocol. In this way, the static route “**floats**” and **is not used** when the route with the better administrative distance is active.

Configure Floating Static Routes

Configure IPv4 and IPv6 Floating Static Routes

The commands to configure default and floating IP default routes are as follows:

```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.2.2
R1(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.2 5
R1(config)# ipv6 route ::/0 2001:db8:acad:2::2
R1(config)# ipv6 route ::/0 2001:db8:feed:10::2 5
```



The **show ip route** and **show ipv6 route** output verifies that the default routes to R2 are installed in the routing table. Note that the IPv4 floating static route to R3 is not present in the routing table.

```
R1# show ip route static | begin Gateway
Gateway of last resort is 172.16.2.2 to network 0.0.0.0

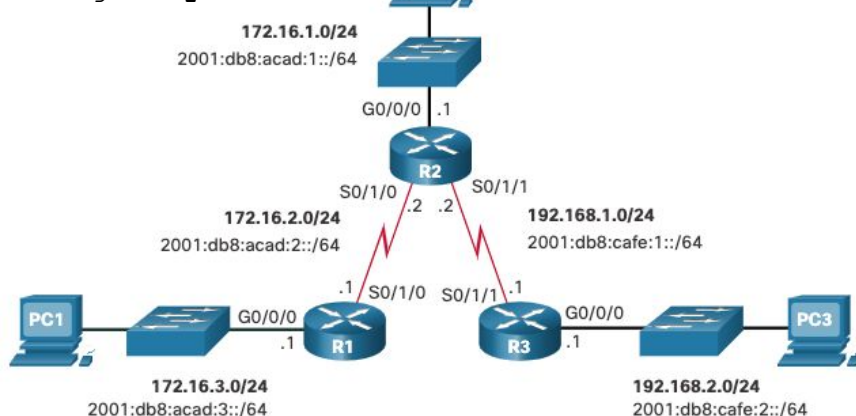
S*    0.0.0.0/0 [1/0] via 172.16.2.2
R1# show ipv6 route static | begin S :
S     ::/0 [1/0]
      via 2001:DB8:ACAD:2::2
R1#
```

Configure Floating Static Routes

Test the Floating Static Routes

- What would happen if R2 failed? To simulate this, R2 **shuts down** both of its **serial interfaces**.
- R1 automatically generates syslog messages for the link going down.
- A look at R1's routing table would show **the secondary route** being used.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.2.2
R1(config)# ip route 0.0.0.0 0.0.0.0 10.10.10.2 5
R1(config)# ipv6 route ::/0 2001:db8:acad:2::2
R1(config)# ipv6 route ::/0 2001:db8:feed:10::2 5
```



```
R1# show ip route static | begin Gateway
Gateway of last resort is 10.10.10.2 to network 0.0.0.0
S*   0.0.0.0/0 [5/0] via 10.10.10.2
R1# show ipv6 route static | begin ::
S    ::/0 [5/0]
      via 2001:DB8:FEED:10::2
R1#
```

Summary static route

Summarization helps to reduce the number of routes in the routing table, minimizing route updates, reduces memory and CPU utilization, and faster routing table lookups.

Multiple static routes can be summarized into a single static route if the following conditions are met:

- The destination networks are in a continuous of networks and can be summarized into a single network address.
- All of the multiple static routes use the same exit interface or next-hop IP address.
- Example to calculate route summarization for the following networks:

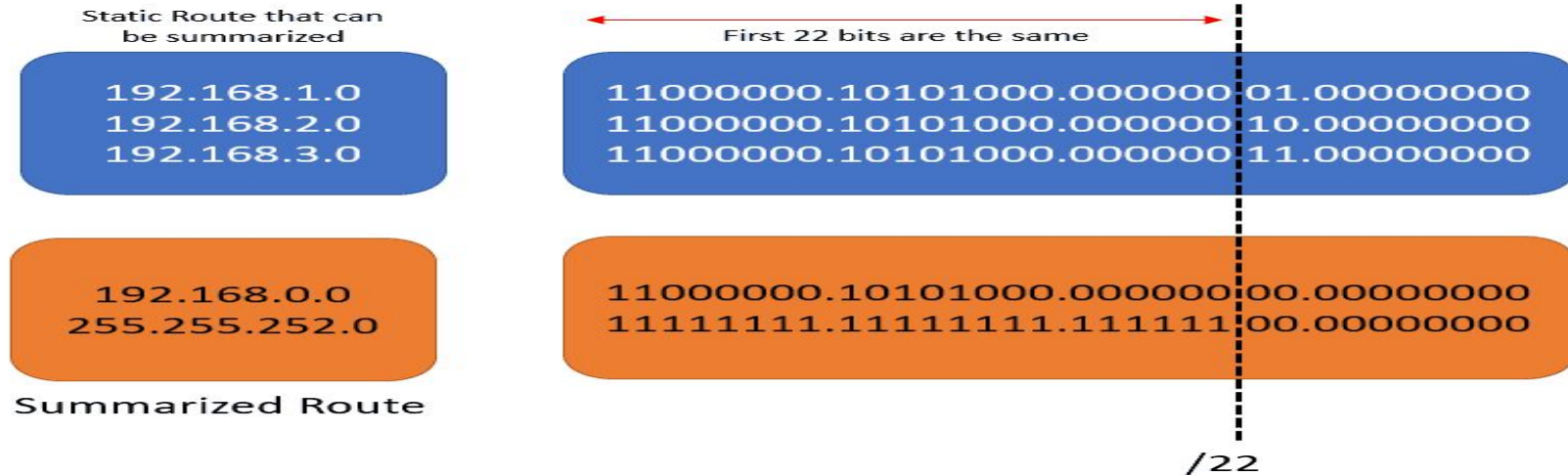
```
R1# show ip route static | begin Gateway Gateway of last resort is not set
192.168.0.0/24 is subnetted, 3 subnets
```

```
S 192.168.1.0 is directly connected, Serial0/2/1
S 192.168.2.0 is directly connected, Serial0/2/1
S 192.168.3.0 is directly connected, Serial0/2/1
```

Summary static route

Steps to calculate the networks for route summarization:

1. Write out the network addresses(192.168.1.0, 192.168.2.0, 192.168.3.0) to be summarized in binary.
2. Find the **subnet mask for summarization**. Start with the far left bit, going to the right, finding all the bits that match until we reach the unmatched column. This signifies the summary boundary.
3. Count the number of far-left matching bits. In this example, it is 22. This number identifies the subnet mask, in slash notation, for the summarized route. **Subnet mask is /22 or a 255.255.252.0** in dotted decimal form.
4. To find the **network address for summarization**, copy the matching 22 bits and add all 0 bits to the end to make 32 bits. **Network address is 192.168.0.0**.
5. Summary static route is ip route 192.168.0.0 255.255.252.0 s0/2/1



15.5 Configure Static Host Routes

Configure Static Host Routes

Host Routes

A **host route** is an IPv4 address with a **32-bit mask**, or an IPv6 address with a **128-bit mask**. The following shows the three ways a host route can be added to the routing table:

- a) **Automatically** installed when an IP address is configured on the router
- b) Configured as a **static host route**
- c) Host route automatically obtained through other methods (discussed in later courses)

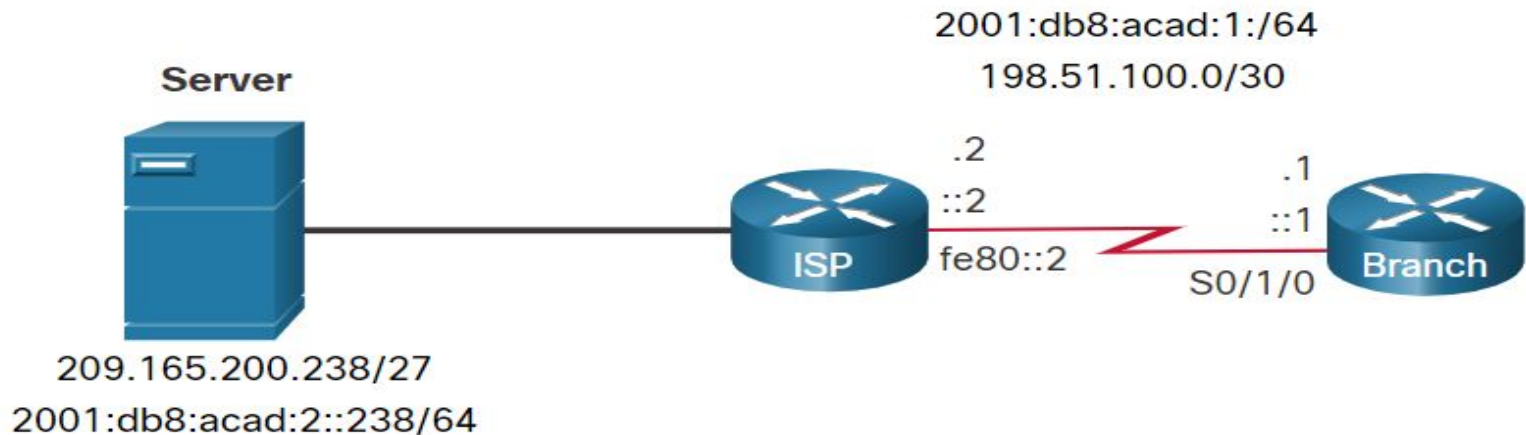
Configure Static Host Routes

Automatically Installed Host Routes

Cisco IOS **automatically** installs a **host route**, also known as a **local host route**, when an **interface address** is configured on the router. A host route allows for a more efficient process for packets that are directed to the router itself, rather than for packet forwarding.

This is in addition to the connected route, designated with a **C** in the routing table for the network address of the interface.

The **local routes** are marked with **L** in the output of the routing table.



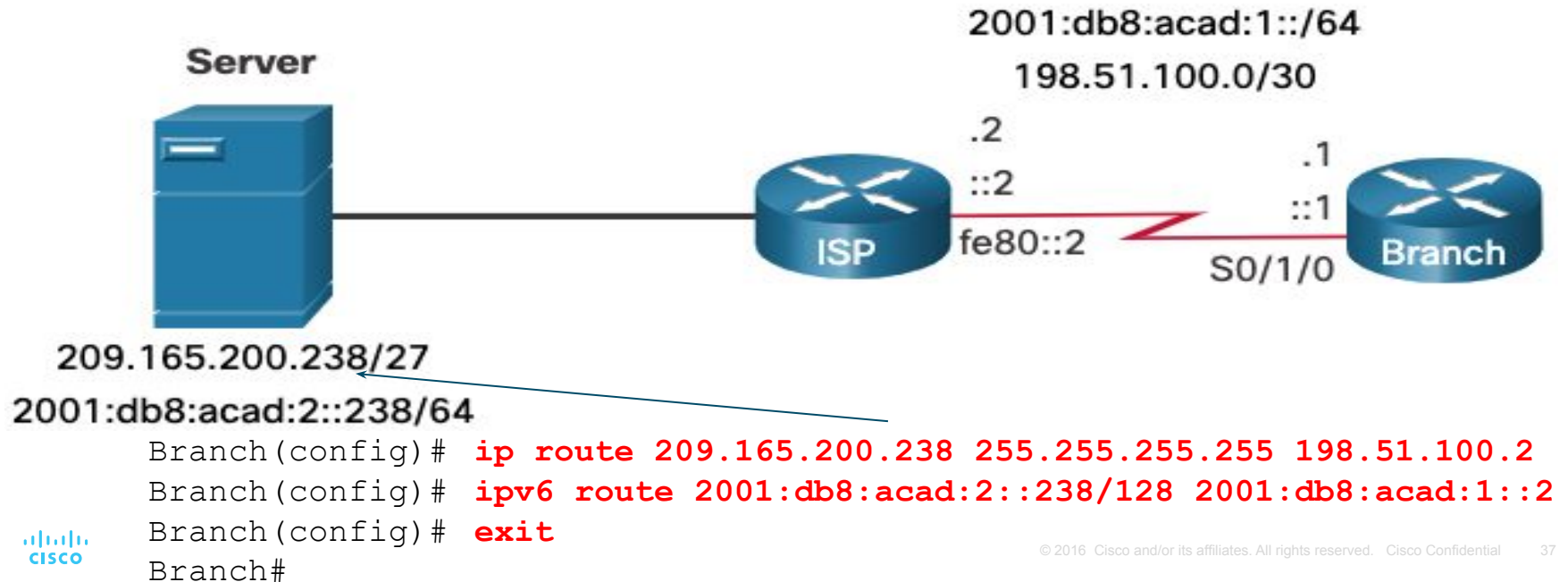
```
Branch# show ip route | begin Gateway
Gateway of last resort is not set
    198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, Serial0/1/0
L       198.51.100.1/32 is directly connected, Serial0/1/0
Branch# show ipv6 route | begin ::
C  2001:DB8:ACAD:1::/64 [0/0]
    via Serial0/1/0, directly connected
L  2001:DB8:ACAD:1::1/128 [0/0]
    via Serial0/1/0, receive
L  FF00::/8 [0/0]
    via Null0, receive
```

a **host route**, also
known as a **local
host route**

Configure Static Host Routes

Static Host Routes

A host route can be a **manually configured** static route to direct traffic to a specific destination device, such as the **server** shown in the figure. The static route uses a **destination IP address** and a **255.255.255.255 (/32)** mask for IPv4 host routes, and a /128 prefix length for IPv6 host routes.



Configure Static Host Routes

Configure Static Host Routes

The example shows the IPv4 and IPv6 static host route configuration on the Branch router to access the server.

```
Branch(config)# ip route 209.165.200.238 255.255.255.255 198.51.100.2  
Branch(config)# ipv6 route 2001:db8:acad:2::238/128 2001:db8:acad:1::2  
Branch(config)# exit  
Branch#
```

Configure Static Host Routes

Verify Static Host Routes

A review of both the IPv4 and IPv6 route tables verifies that the routes are active.

```
Branch# show ip route | begin Gateway
Gateway of last resort is not set
    198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, Serial0/1/0
L       198.51.100.1/32 is directly connected, Serial0/1/0
    209.165.200.0/32 is subnetted, 1 subnets
S       209.165.200.238 [1/0] via 198.51.100.2
Branch# show ipv6 route
(Output omitted)
C    2001:DB8:ACAD:1::/64 [0/0]
    via Serial0/1/0, directly connected
L    2001:DB8:ACAD:1::1/128 [0/0]
    via Serial0/1/0, receive
S    2001:DB8:ACAD:2::238/128 [1/0]
    via 2001:DB8:ACAD:1::2
Branch#
```

Configure IPv6 Static Host Route with Link-Local Next-Hop

For IPv6 static routes, the **next-hop address** can be the **link-local address** of the adjacent router. However, you must specify an **interface type** and **an interface number** when using a **link-local address** as the next hop, as shown in the example.

First, the original IPv6 static host route is **removed**, then a fully specified route configured with the IPv6 address of the server and the IPv6 link-local address of the ISP router.

```
Branch(config)# no ipv6 route 2001:db8:acad:2::238/128 2001:db8:acad:1::2
Branch(config)# ipv6 route 2001:db8:acad:2::238/128 serial 0/1/0 fe80::2
Branch# show ipv6 route | begin ::
C   2001:DB8:ACAD:1::/64 [0/0]
    via Serial0/1/0, directly connected
L   2001:DB8:ACAD:1::1/128 [0/0]
    via Serial0/1/0, receive
S   2001:DB8:ACAD:2::238/128 [1/0]
    via FE80::2, Serial0/1/0
Branch#
```

link-local address

interface type and an interface number

15.6 Module Practice and Quiz

Packet Tracer – Configure IPv4 and IPv6 Static and Default Routes

In this Packet Tracer, you will do the following:

- Configure IPv4 Static and floating static default routers

- Configure IPv6 static and floating static default routes

- Configure IPv4 static and floating static routes to internal LANs

- Configure IPv6 static and floating static routes to the internal LANs

- Configure IPv4 host routes

- Configure IPv6 host routes

Lab - Configure IPv4 and IPv6 Static and Default Routes

In this lab, you will complete the following objectives:

- Build the Network and Configure Basic Device Settings

- Configure and Verify IP and IPv6 Addressing on R1 and R2

- Configure and Verify Static and Default Routing for IPv4 on R1 and R2

- Configure and Verify Static and Default Routing for IPv6 on R1 and R2

What Did I Learn In This Module?

- Static routes can be configured for IPv4 and IPv6. Both protocols support the following types of static routes: standard static route, default static route, floating static route, and summary static route.
- When configuring a static route, the next hop can be identified by an IP address, exit interface, or both. How the destination is specified creates one of the three following types of static route: next-hop, directly connected, and fully specified.
- IPv4 static routes are configured using the following global configuration command: `ip route network-address subnet-mask { ip-address | exit-intf [ip=address] } [distance]`.
- IPv6 static routes are configured using the following global configuration command: `ipv6 route ipv6-prefix/prefix-length { ipv6-address | exit-intf [ipv6-address]} [distance]`.
- In a next-hop static route, only the next-hop IP address is specified. The exit interface is derived from the next hop.
- When configuring a static route, another option is to use the exit interface to specify the next-hop address. Directly connected static routes should only be used with point-to-point serial interfaces.
- In a fully specified static route, both the exit interface and the next-hop IP address are specified. This form of static route is used when the exit interface is a multi-access interface and it is necessary to explicitly identify the next hop. The next hop must be directly connected to the specified exit interface.
- In a fully specified IPv6 static route, both the exit interface and the next-hop IPv6 address are specified.

What Did I Learn In This Module? (Cont.)

- A default route is a static route that matches all packets.
- Default static routes are commonly used when connecting an edge router to a service provider network, and a stub router.
- The command syntax for an IPv4 default static route is similar to any other IPv4 static route, except that the network address is 0.0.0.0 and the subnet mask is 0.0.0.0.
- The command syntax for an IPv6 default static route is similar to any other IPv6 static route, except that the ipv6-prefix/prefix-length is ::/0, which matches all routes.
- Floating static routes are static routes that are used to provide a backup path to a primary static or dynamic route in the event of a link failure.
- The floating static route is configured with a higher administrative distance than the primary route. By default, static routes have an administrative distance of 1, making them preferable to routes learned from dynamic routing protocols.
- IP floating static routes are configured by using the distance argument to specify an administrative distance.
- A host route is an IPv4 address with a 32-bit mask or an IPv6 address with a 128-bit mask.

What Did I Learn In This Module? (Cont.)

- There are three ways a host route can be added to the routing table: automatically installed when an IP address is configured on the router, configured as a static host route, or automatically obtained through other methods not covered in this module.
- Cisco IOS automatically installs a host route, also known as a local host route, when an interface address is configured on the router.
- A host route can be a manually configured static route to direct traffic to a specific destination device.
- For IPv6 static routes, the next-hop address can be the link-local address of the adjacent router; however, you must specify an interface type and an interface number when using a link-local address as the next hop. To do this, the original IPv6 static host route is removed, then a fully specified route is configured with the IPv6 address of the server and the IPv6 link-local address of the ISP router.

New Terms and Commands

- static route
- default static route
- floating static route
- summary static route
- next-hop route
- directly connected static route
- Fully specified static route
- `ip route network-address subnet-mask { ip-address | exit-intf [ip-address]} [distance]`
- `ipv6 route ipv6-prefix/prefix-length {ipv6-address | exit-intf [ipv6-address]} [distance]`
- `show ip route static`
- `show ipv6 route static`
- quad-zero route
- `ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf}`
- `ipv6 route ::/0 {ipv6-address | exit-intf}`
- host route
- local host route
- static host route

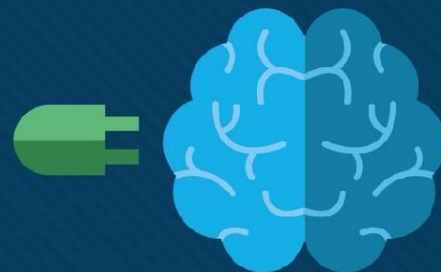




Chapter 1 Part 2

Module 16: Troubleshoot Static and Default Routes

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: Troubleshoot Static and Default Routes

Module Objective: Troubleshoot static and default route configurations.

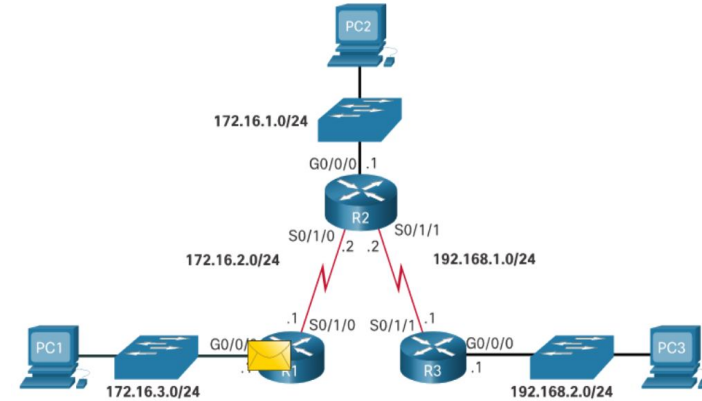
Topic Title	Topic Objective
Packet Processing with Static Routes	Explain how a router processes packets when a static route is configured.
Troubleshoot IPv4 Static and Default Route Configuration	Troubleshoot common static and default route configuration issues.

16.1 Packet Processing with Static Routes

Packet Processing with Static Routes

Static Routes and Packet Forwarding

- PC1 addresses a packet to PC3 and sends it to the default gateway address.
- When the packet arrives on the R1 G0/0/0 interface, R1 **decapsulates** the packet and **searches the routing table** for a matching destination network entry.



If the destination IP address:

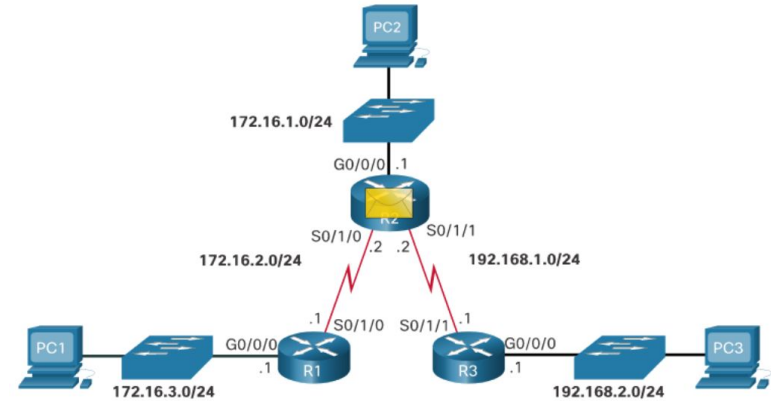
- **Matches a static route entry**, R1 will use the static route to identify the **next-hop IP address** or **exit interface**.
- **Does not match** a specific route to the destination network, then R1 will use the **default static route** (if configured).
- **Does not match** a route table entry, then R1 will **drop** the packet and **send an ICMP message back** to the source (i.e., PC1).

Packet Processing with Static Routes

Static Routes and Packet Forwarding (Cont.)

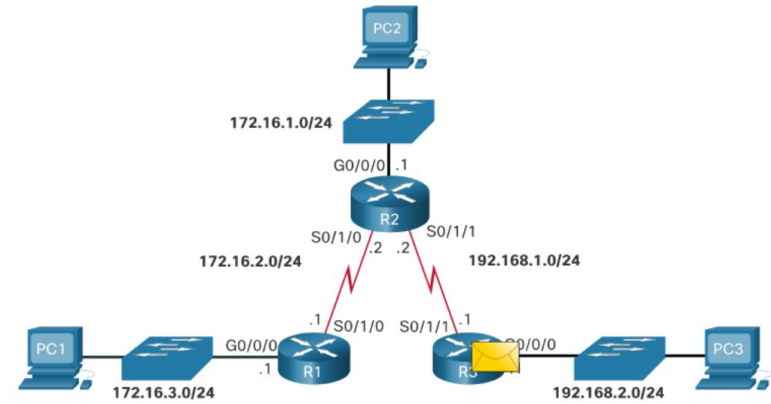
Assuming **R1** **matched** a routing table entry, it encapsulates the packet in a new frame and **forwards** it out of interface **S0/1/0** to **R2**.

- **R2** receives the packet on its **S0/1/0** interface.
- It decapsulates and processes the packet the same way R1 did.
- When R2 finds a **match** in the routing table, it uses the identified **next-hop IP address** or **exit interface** and sends the packet out of its **interface S0/1/1** towards R3.



Static Routes and Packet Forwarding (Cont.)

- **R3** receives the packet, decapsulates it, and searches the routing table for a match.
- The **destination IP address** of PC3 matches the directly connected **G0/0/0 interface**. Therefore, R3 searches the ARP table for the Layer 2 MAC address of PC3.
- If **no** ARP entry exists, then R3 sends an ARP request out of the **G0/0/0 interface**.



- PC3 responds with an ARP reply containing its MAC address.
- R3 **encapsulates** the packet in a new frame and uses the PC3 MAC address as the destination MAC address and the G0/0/0 MAC address as the source MAC address.
- The frame is forwarded out of **interface G0/0/0** and PC3 receives and processes it accordingly.

16.2 Troubleshoot IPv4 Static and Default Route Configuration

Troubleshoot IPv4 Static and Default Route Configuration

Network Changes

Networks fail for a number of reasons:

- An **interface** can **fail**

- A service provider **drops** a connection

- Links can become **oversaturated**

- An administrator may enter a **wrong** configuration.

Network administrators are responsible for pinpointing and solving the problem.

To efficiently find and solve these issues, it is advantageous to be intimately familiar with tools to help isolate routing problems quickly.

Troubleshoot IPv4 Static and Default Route Configuration

Common Troubleshooting Commands

Command	Description
<code>ping</code>	• Verify Layer 3 connectivity to destination. • Extended pings provide additional options.
<code>tracert</code>	• Verify path to destination network. • It uses ICMP echo reply messages to determine the hops to the destination.
<code>show ip route</code>	• Displays the routing table. • Used to verify route entries for destination IP addresses.
<code>show ip interface brief</code>	• Displays the status of device interfaces. • Used to verify the operational status and IP address of an interface.
<code>show cdp neighbors</code>	• Displays a list of directly connected Cisco devices. • Also used to validate Layer 1 and 2 connectivity.

Troubleshoot IPv4 Static and Default Route Configuration

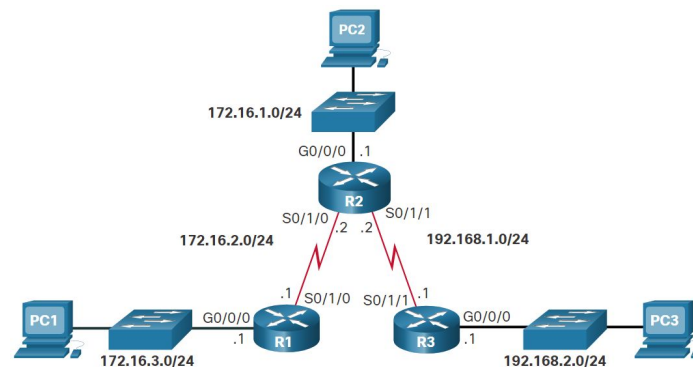
Solve a Connectivity Problem

Connectivity from PC1 to PC3 fails.

Extended pings from the R1 G0/0/0 interface to PC3 fail.

Pings from R1 (i.e., S0/1/0 interface) to R2 are successful.

Pings from R1 (i.e., S0/1/0 interface) to R3 are successful.



- R2 routing table reveals the problem and the **incorrect static route** is removed.
- A new static route solves the problem.
- `ip route 172.16.3.0 255.255.255.0 172.16.2.1`

```
R2# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       172.16.1.0/24 is directly connected,
        GigabitEthernet0/0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0/0
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.2/32 is directly connected, Serial0/1/0
S       172.16.3.0/24 [1/0] via 192.168.1.1
        192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Serial0/1/1
L       192.168.1.2/32 is directly connected, Serial0/1/1
S       192.168.2.0/24 [1/0] via 192.168.1.1
```

R2#

16.3 Module Practice and Quiz

Packet Tracer – Troubleshoot Static and Default Routes

In this activity you will troubleshoot and static and default routes and repair any errors that you find.

- Troubleshoot IPv4 static routes.
- Troubleshoot IPv6 static routes.
- Configure IPv4 static routes.
- Configure IPv4 default routes.
- Configure IPv6 static routes.

Structured Design

Lab - Troubleshoot Static and Default Routes

In this lab, you will complete the following objectives:

- Evaluate Network Operation.
- Gather information, create an action plan, and implement corrections.

What did I learn in this module?

- A host sends a packet to another host and sends it to the default gateway address.
- When the packet arrives on a router interface, it decapsulates the packet and searches the routing table for a matching destination network entry.
- If the destination IP address:
 - Matches a static route entry, the router will use the static route to identify the next hop IP address or exit interface.
 - Does not match a specific route to the destination network, then the router will use the default static route (if configured).
 - Does not match a route table entry, then the router will drop the packet and send an ICMP message back to the source.
- If the router matched a routing table entry, then the router encapsulates the packet and forwards it out of the appropriate interface.
- The packet is forwarded from router to router until it reaches its destination network.

What did I learn in this module? (Cont.)

- When the packet reaches the destination network, that router will search the routing table for a match.
- When the destination IP address matches a directly connected Ethernet interface, the router searches the ARP table for the Layer 2 MAC address of the destination IP address.
- If no ARP entry exists, the router sends an ARP request out of the Ethernet interface
- The destination host responds with an ARP reply containing its MAC address.
- The router then encapsulates the packet in a new frame. It uses the MAC address of the destination host as the frame destination MAC address, and the MAC address of the router Ethernet interface as the source MAC address in the frame.
- The frame is forwarded out of the appropriate interface.
- The packet arrives on the network interface card (NIC) interface of destination host and is processed accordingly.

What did I learn in this module? (Cont.)

Common IOS troubleshooting commands to troubleshoot IPv4 Static and default routes include:

- **ping**
- **tracert**
- **show ip route**
- **show ip interface brief**
- **show cdp neighbors detail**

